YASH Technologies

More than what you think.

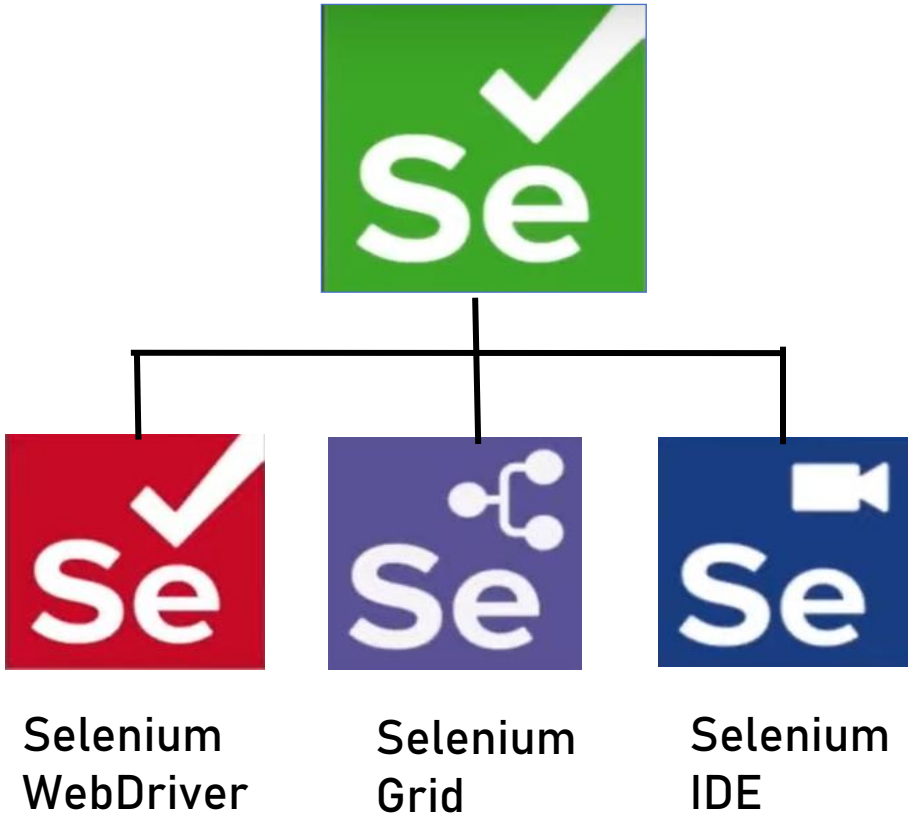
Selenium 4 Architecture, new features and Improvements

Agenda

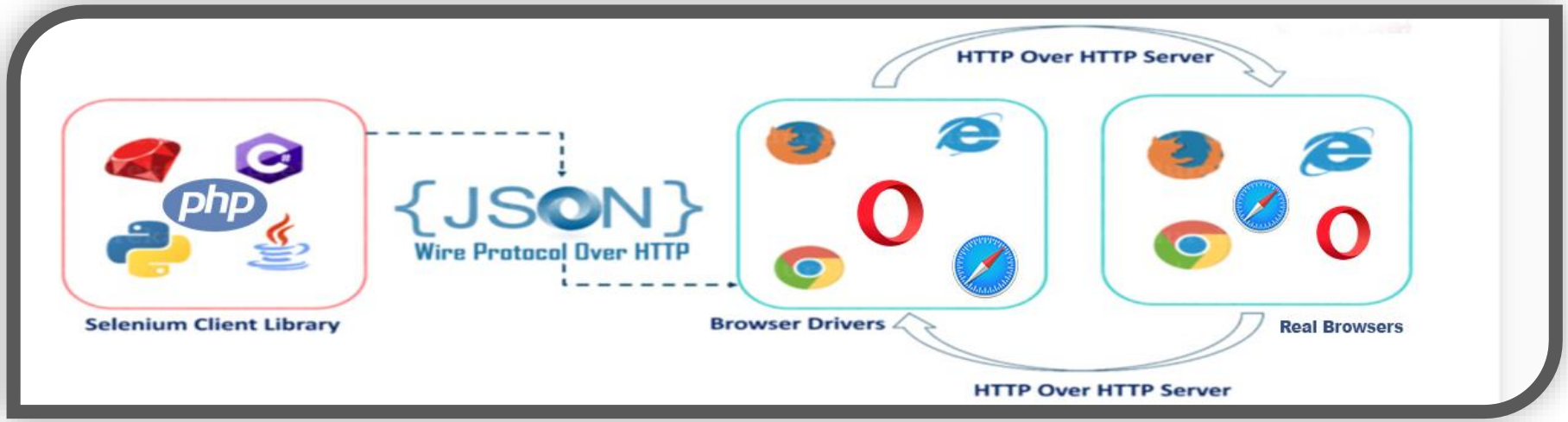
- Selenium 4 Architecture
- Selenium 4 Components
- Enhancements with Selenium 4.10.0
  - Moving to higher version of Java (Java 11 Or 17)
  - No need to manually setup Browser Drivers
  - Relative Locators
  - New Window/Tab Management
  - New Selenium Grid
  - DesiredCapabilities Modification
  - Action Class Changes

Selenium 4 Components

- Selenium 4 was actually introduced in 2018 with alpha and beta versions with some changes and new features. The first stable version was introduced in 2021.
- Now selenium 4 is much more stable with version 4.10.0 and its high time to switch from selenium 3 to selenium 4.
- Selenium 4 has 3 components and all 3 components are using W3C standards.
  - Selenium WebDriver – used to write scripts for automating browsers.
  - Selenium Grid – helps in performing parallel execution on standalone and distributed environments on different OS and browsers.
  - Selenium IDE – It's a record and playback tool available as a add-on on Chrome, Firefox and MSEdge



Selenium 3 Architecture



- The flow of data transfer happens over four components:
- Selenium Client Libraries/Language Bindings is available in many languages like java, ruby, , python, JavaScript, php etc
- JSON Wire Protocol over HTTP Responsible to understand the command issued by selenium language binding and convert it into JSON
- Browser Drivers When a browser driver receives a JSON request, it will be sent to the respective browser over HTTP server and the HTTP response will go back in the form of JSON
- Real Browsers Receives the request from respective browser driver and performs an action.

Selenium 4 Architecture

- The main drawback of JSON wire protocol was that there was no direct communication between the client libraries and the browser driver. JSON wire protocol acts as a mediator between client libraries and Webdriver.
- The server in browser driver has nothing to do with the programming code but only understands protocols, hence it uses the process of serialization (convert object's data to JSON format) and deserialization (convert JSON format to object). This results sometime in exceptions, slower test execution, and more chances of the unstable tests.
- As almost all real browsers and their driver executables are already on W3C standards and now client libraries or language bindings also uses the W3C protocol making communication much easier between client and server (driver). It is like two same language-speaking persons talking to each other. W3C protocol is actually included in language binding.
- Since selenium version 3.8 both json wire and w3c protocol were there in the web driver. But since version 4.0 selenium is only using World Wide Web Consortium (W3C) standard.
- Also from selenium version 4.6 upwards, there is no need to manually setup browser driver in your machine because it is automatically taken care of by SeleniumManager included in client library.
- All the browsers and client library have adopted the w3c protocol which makes Selenium codebase more optimized.

Moving to higher version of Java (Java 11 or 17)

- After September 2023, selenium will stop supporting Java version 8.
- Now the newer minimum java version, selenium will support is java 11.
- But its not confirmed, they might move to java 17.
- According to post on their official website:

> Selenium has long supported Java 8, but as technology evolves, so must we. One of the primary reasons for this change is that Java 8 reached the end of active support over a year ago. In addition, our default HTTP Client has not had a major release in several years, and a bug has been found that we can not fix. We have decided to move to the native Java HTTP Client, but it requires using Java 11 or greater. The sooner we make this change, the sooner we can avoid dealing with this issue.
>
> Our new minimum version will be Java 11. September 30, 2023 is also the end of active support for Java 11. However, we want to take a cautious and conservative path forward, and not force our users to make the big jump from Java 8 to Java 17, as we understand the community might need longer to move to that version. We will revisit this topic in the future to announce the plan to support Java 17 as a minimum version.

- I have already moved to Java 17 to work with selenium 4 and there is not much change.

No need to manually setup Browser Drivers

- With selenium version 4.6 and upwards, there is no need to set property for browser driver and no need to manually place them in your system or set their environment path. Even we do not need WebDriverManager from bonigarcia.
- Because now Webdriver API contains SeleniumManager class which is a singleton class again written by bonigarcia.
- SeleniumManager class verifies that driver executable for the specific browser version is there in the system or otherwise downloads it for us and we don't have to take care. And communication between WebDriver and browser takes place in a same way ie. via browserdriver.
- SeleniumManager implementation is still in beta ie. they have not finalized it yet, and may change.
- SeleniumManager class downloads the browser executable in your system at the location:
- \Users\<your_name>\.cache\selenium\
- According to selenium's official website:

Selenium: now with batteries included!

The Selenium project wants to improve the user experience, and one of the first steps is to help all users to simplify how they set up their environment. Configuring browser drivers has been for many years a task which users need to perform in order to have a working environment to run Selenium.

Setting up a browser driver once is not that complicated, but as browser release cycles got shorter, and now we have new Chrome/Firefox/Edge version every 4-6 weeks, the task of keeping the browser driver in sync with the browser version is not that easy anymore.

Selenium Manager is a new tool that helps to get a working environment to run Selenium out of the box. Beta 1 of Selenium Manager will configure the browser drivers for Chrome, Firefox, and Edge if they are not present on the `PATH`.

Relative Locators

- By class provides a mechanism by which to find elements within the HTML document.
- By class is known as Locator class because in selenium3, it has mainly 8 Location Strategies for finding element.
- Location strategies linkText and partialLinkText can only be used in case of hyperlink <a> tag
- In addition to these location strategies, selenium4 has some relative locators which are methods to locate nearby elements by specifying directions.

Current Location strategies of By class:	New Relative Locators in Selenium 4
Id name Class Name XPath CssSelector linkText partialLinkText tagName	above below toLeftOf toRightOf near

8

New Window/Tab Management

- Earlier we could not open new window or tabs in the same session of webdriver, although we had javascript for doing the same but we had to achieve this using JavascriptExecutor:
  - *((JavascriptExecutor)driver).executeScript("window.open('http://google.com1, '_blank');");* // opening new tab using JavascriptExecutor
- Now Webdriver API has method to work with multiple tabs/windows in the same session without creating new WebDriver object.

```
driver.switchTo().newWindow(WindowType.WINDOW);
```

Open and switch to new window

```
driver.switchTo().newWindow(WindowType.TAB);
```

Open and switch to new tab

Selenium Grid

- Selenium Grid aims to run tests on several machines/OS and several browsers by distributing them (parallel execution)
- In selenium 4, selenium grid is completely redesigned and code is completely new ie. from scratch.
- In Selenium4, new grid can be used in three ways.
  - Standalone – This mode starts a fully functional Grid with a single command and on a single machine.
  - Hub and Node - Hub and Node is the most used role because it allows to combines different machines/OS or environments in a single Grid. Its use is exactly the same as was in Selenium 3.
  - Distributed mode – This mode starts each component separately. Grid now has inbuilt Docker support and there is no need to create new VMs.

DesiredCapabilities Modification and Actions Class Changes

- DesiredCapabilities class in WebDriver API is a series of key/value pairs that stores the browser properties like browser name, browser version, the path of the browser driver in the system, supports JavaScript or not etc. to determine the behaviour of the browser at a run time while executing selenium grid.
- In selenium 4, this class is replaced with Browser Options Class. So, there is separate Options class for all the browsers like ChromeOptions, FirefoxOptions, EdgeOptions, InternetExplorerOptions, etc.
- The Actions class is a API for emulating complex *keyboard and mouse actions in Selenium Webdriver* (like double click, right click, mouse hover, etc).
- In selenium 4, Actions API has included some newer methods to perform certain tasks easily like zoom in/zoom out, two key press etc.